

VICTOR OSE IYAYI

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EXPERIENCE

TURBHAM TECHNOLOGIES

Backend Engineer/DevOps Engineer

Abuja, Nigeria

September 2023 —

1) Developed a Comprehensive API that serves the web app and mobile app

I architected and developed HealthBook, a comprehensive health monitoring ecosystem designed to optimise clinical workflows and enhance the patient-provider experience. I engineered a high-performance Python API to serve as a unified backbone for both mobile and web platforms, ensuring seamless data synchronisation across the entire system.

To ensure scalability and reliability, I orchestrated a multi-service Docker infrastructure, containerising the API alongside PostgreSQL for relational data integrity and Redis for high-speed caching. I managed the end-to-end deployment on a Linode server, utilising container orchestration to maintain a stable, production-ready environment that bridged the gap between complex medical data and intuitive user accessibility.

- Key accomplishments of the project include:

- **User Authentication:** I built a secure user authentication system to safeguard user data. This system enables users to register, log in, and manage their passwords with confidence. To further strengthen security, I implemented two-factor authentication (2FA) with recovery codes, providing an extra layer of protection for user accounts. Additionally, the system automatically generates unique usernames, ensuring a seamless signup experience.
- **CI/CD:** Streamlined production delivery through a GitHub Actions CI/CD workflow utilising a Blue-Green strategy; engineered a seamless environment-switching mechanism to eliminate deployment-related downtime and enable instant rollbacks.
- **Database Integration:** Utilised Postgres as the database management system for efficient storage and retrieval of all data, while storing raw files like images and documents in the Postgres docker volume. Utilised Redis as the session and cache management system, storing the values in the Redis docker volume. Volumes were used for storage so as to persist data in case of container loss or downtime.
- **Domain management:** I was able to manage the domain of the API with a Content Delivery Network, CloudFlare, using Nginx as my Reverse Proxy service on the server, I was able to route traffic from CloudFlare to the server where the API is hosted.
- **Patient Appointment Management:** I built a user-friendly CRUD (Create, Read, Update, Delete) appointment booking system. Patients can easily schedule appointments with 20-minute intervals, receiving instant email confirmations for bookings, updates, and cancellations.
- **Secure Document Management:** A robust CRUD document upload system allows patients and medical staff to securely upload and share medical documents in PDF format. Additionally, patients can download a history of their medical records within a

specific date range for easy access.

- **Connections:** A unique friend request system fosters a support network. Patients can connect with family and friends, allowing authorised users to view vitals and even book appointments on their behalf, promoting collaborative care.
- **Flexibility and Choice:** A many-to-many relationship between users, hospitals, and pharmacies provides patients with the freedom to choose from multiple healthcare providers within the system.
- **Healthcare Network:** Hospitals can leverage the health centre linking system to connect with patients and request permission to access their medical records. This fosters better communication and care.
- **Patient Onboarding:** Hospitals can streamline patient registration through a dedicated signup system, ensuring a smooth start to their healthcare journey, under the specified hospital.
- **Hospital Appointment Management:** The system facilitates appointment scheduling for patients linked to the hospital. Hospitals receive notifications for bookings, updates, and cancellations, ensuring everyone stays on the same page.
- **Hospital Document Sharing:** Both patients and medical staff can securely upload and access medical documents (PDF format) for patients under the hospital's care, promoting informed decision-making.
- **Doctor Profiles:** Medical staff can manage their profiles within the system, update their personal information and licenses, fostering professionalism and transparency for patients.
- **Flexible Staffing Solutions:** The system allows doctors and medical staff to be associated with multiple health centres, providing them with greater flexibility and hospitals with access to a wider talent pool.
- **Device Management:** A user-friendly device management system allows for easy linking and unlinking of patients to their 4G or Bluetooth devices, streamlining data collection and monitoring.
- **Content Management For Admins:** A user-friendly blog management system empowers admins to create, edit, and delete blog posts, keeping users informed and engaged.
- **Device Integration:** Admins have the ability to register devices within the system, paving the way for vital data collection and real-time health monitoring.

Awari
Backend Engineer

Lagos, Nigeria
February 2025 –

2) Maintaining Awari's App and Web Services

Leveraged deep insights into user behaviour and journey mapping gained from my background in health-tech to architect Awari's dual-sided payment ecosystem. By prioritising the end-to-end user experience, I engineered robust financial tracking tools for consumers while streamlining seamless payment acquisition for merchants, ensuring a frictionless transition from discovery to transaction.

- Key accomplishments of the project include:

- **Executed High-Stakes Database Migration:** To support multi-factor and social authentication (Google, Facebook) with zero service interruption for existing users.
- **Optimised Revenue Retention:** By implementing a flexible subscription logic that allows businesses to renew plans prior to expiration, improving cash flow predictability
- **Harmonised Fragmented Database Systems:** Within the first week of tenure, significantly reducing technical debt and improving query performance across the platform.
- **Enhanced User Engagement:** By building a multi-provider calendar integration (Google, Apple, Yahoo) and an automated email reminder service for event attendees.
- **Engineered Unified identity System:** By allowing users to seamlessly link and manage dual Vendor/Customer profiles under a single account.
- **Facilitated Ecosystem Growth:** Through strategic 3rd-party integrations, including a seamless registration bridge between Awari and Reisty accounts.

Etranzact Global
Backend Engineer

Lagos, Nigeria
March 2025 –

3) Maintaining Etranzact's App Service

Utilised my skills and expertise to build and implement extensive user and merchant payments systems. Keeping track of user's finances while empowering merchants to receive payments for their products seamlessly.

- Key accomplishments of the project include:

- **MOMO Payment:** I integrated Etranzact's payment system with the mobile money payment in Ghana to facilitate bulk payment in one transaction.
- **Budgeting System:** I created the current budgeting system for Etranzact's App from scratch to production, used by thousands of users. This allows users track their spendings relative to their set budget.
- **Compliance:** I created the account limit feature for users of the ussd service on Etranzact Ghana in accordance to governmental guidance.
- **USSD Service:** I created the entire flow for merchants to apply for Etranzact's USSD service code, process the application and assign the merchant a USSD short code.
- **Sierra Leone Payments:** I integrated Sierra Leone's payment service from the app, integrating the service provider in the country for Momo and bank payments.
- **Bulk MOMO PAYMENT:** Designed and implemented a bulk MoMo payment solution that aggregated total funds for centralised funding before distributing individual proceeds to recipients, significantly scaling transaction throughput.

Maintaining the backend of Etranzact's global system has opened my mind to building scalable applications that can survive the mass onslaught of user payments and merchant management.

EDUCATION

ALT SCHOOL AFRICA

Diploma in Software Development

Lagos, Nigeria

June 2023

- **Diploma in Software Development (Backend):** Successfully completed a comprehensive diploma program focused on backend development, acquiring proficiency in technologies such as Flask, Django, Django Rest Framework, SQLite, Postgres, Flask-RestX, Flask-Smorest, ApiFlask, Jinja2, Git, Github, Python, Open Source, and Version Control.
- **Proficient in Flask and Django:** Demonstrated expertise in utilising Flask and Django frameworks to develop robust and scalable backend solutions, including RESTful APIs and web applications.
- **Database Management:** Skilled in working with SQLite and Postgres, for efficient data storage and retrieval, ensuring seamless integration with backend systems.
- **API Development:** Developed RESTful APIs using Flask-RestX and Flask-Smorest, enabling smooth communication between frontend and backend systems.
- **Version Control and Collaboration:** Proficient in Git and GitHub for effective version control and collaborative development, ensuring a streamlined workflow and efficient teamwork.
- **Frontend Proficiency:** Knowledgeable in HTML and CSS, enabling effective collaboration with frontend developers and contributing to the overall user experience.
- **Technical Writing:** Possess strong technical writing skills, allowing for clear and concise documentation of code, project specifications, and system architecture.
- **Successfully Deployed Multiple APIs:** Demonstrated proficiency in deploying several APIs, showcasing the ability to implement and launch backend systems to production environments effectively.
- **Personal App Maintenance:** Actively maintaining a personal application, exemplifying a commitment to continuous learning and hands-on experience in managing and improving live projects.

UNIVERSITY OF BENIN

*Bachelor of Arts in Religions;
2014*

Benin, Edo State

January 2011 - December

- Bachelor of Science in Religions.
- Minor in History and Philosophy

Technical Skills

Languages: Python, Javascript (NodeJs), HTML/CSS, Typescript, Golang, Java

Frameworks: Flask, RestX, Express, React Native, Django, Flask Restful, Springboot, NestJs, Express

Technologies: Postgres, Shell scripts, Linode (Akamai), CloudFlare, SQL operations, API documentation (Swagger), MongoDB, Docker, BLE (Bluetooth Low Energy), Sqlite,

Docker volumes, Redis, SQLAlchemy ORM, Cloud based operations

Software: Illustrator, Insomnia, Photoshop, Postman, VS Code

Languages

Fluent in English

Certifications & Training

Human Relationship Management (HRM), Health Safety and Environment (HSE1-2), Customer Relationship Management (CRM), Fundamentals of Digital Marketing (Google), Google ads display certification, Google Analytics